

Sheryar Khan

MERN Stack Developer

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GitHub: <https://github.com/Sheryar7>

Portfolio: <https://portfolio-snowy-six-91.vercel.app/>

SUMMARY

Software Engineering graduate and Full-Stack Developer with hands-on experience building AI-powered web applications, Retrieval-Augmented Generation (RAG) architectures, and scalable microservices using Next.js (App Router), React, Node.js, NestJS, Supabase (pgvector), and MongoDB/PostgreSQL. Experienced in real-world frontend workflows through a recent internship, alongside shipping production-ready SaaS platforms featuring vector similarity search, role-based access control (RBAC), secure JWT/OAuth authentication, and RESTful API architectures. Skilled in integrating LLM embeddings (Gemini API) and leveraging modern AI-assisted development tools (Copilot, ChatGPT, Claude) to accelerate problem-solving and optimize code quality. Seeking a Full-Stack, Web Developer, or MERN internship/junior role to contribute to production-grade systems.

EXPERIENCE

Frontend Developer Intern | Remote

Duration: 2 Months

- Developed modern, dynamic user interfaces for web applications using Next.js, TypeScript, and Tailwind CSS.
- Architected modular, reusable frontend components adhering to clean code standards and responsive design practices.
- Collaborated on client-side feature development, styling UI layouts, and optimizing frontend performance.

PROJECTS

SkillGraph — Candidate RAG(Retrieval-Augmented Generation) & Knowledge Search Engine

🔗 **GitHub:** <https://github.com/Sheryar7/skillgraph>

Description:

Designed and built an AI-powered Retrieval-Augmented Generation (RAG) platform that ingests candidate profile PDFs, extracts and chunks text, converts content into high-dimensional vector embeddings, and executes real-time cosine similarity vector searches.

Key Features:

- Engineered an end-to-end RAG architecture utilizing vector similarity search for semantic candidate querying.
- Built a NestJS microservice backend for asynchronous PDF parsing, text extraction, and recursive chunking.
- Integrated Google Gemini API to generate high-dimensional vector embeddings for profile contextual analysis.
- Implemented Supabase PostgreSQL with `pgvector` extension for efficient vector index storage and cosine similarity scoring (achieving 75%+ precision).
- Developed a Next.js 14 frontend interactive search engine and dashboard with real-time document indexing status updates.

Tech Stack:

Next.js 14, NestJS, TypeScript, Supabase (pgvector), Google Gemini AI API, Tailwind CSS, REST APIs.

SkillForge – Modern LMS SaaS Platform

🔗 **Live Demo:** <https://modern-lms-smoky.vercel.app/>

🔗 **GitHub:** <https://github.com/Sheryar7/skillforge-lms>

Description:

Built and deployed a modern SaaS-style Learning Management System using Next.js App Router, TypeScript, Supabase, and Tailwind CSS with role-based dashboards and scalable architecture.

Key Features:

- Implemented instructor and student role-based dashboards
- Built course curriculum architecture with sections and lessons
- Created protected routes and authentication middleware
- Integrated Supabase authentication and PostgreSQL database

- Designed reusable component-based frontend architecture
- Implemented responsive SaaS-style UI using Tailwind CSS
- Built scalable App Router structure using Server & Client Components
- Added analytics foundation and dashboard architecture
- Deployed production application on Vercel

Tech Stack:

Next.js 15, TypeScript, Supabase, PostgreSQL, Tailwind CSS, React, App Router, Vercel

HMC Time-Off Microservices (NestJS)

🔗 **GitHub:** <https://github.com/Sheryar7/readyon-timeoff-microservice>

Description:

Designed and developed a backend microservice to manage employee time-off requests and maintain balance integrity while simulating synchronization with an external HCM system (like Workday/SAP).

Key Features:

- Built modular backend architecture using NestJS (Controller, Service, Module pattern)
- Designed REST APIs for time-off request creation and retrieval
- Implemented DTO-based validation using class-validator and ValidationPipe
- Integrated SQLite database using TypeORM
- Ensured data consistency with proper request validation and schema alignment
- Handled real-world backend issues like validation errors, database constraints, and data mapping
- Designed system considering real-time and batch sync scenarios with external HCM systems

Tech Stack:

NestJS, TypeScript, SQLite, TypeORM, class-validator

SkillForge – Learning Management System (MERN stack)

🔗 **Frontend:** <https://skillforge-frontend-blue.vercel.app/>

🔗 **GitHub (Frontend):** <https://github.com/Sheryar7/skillforge-frontend>

🔗 **GitHub (Backend):** <https://github.com/Sheryar7/skillforge-backend>

Description:

A full-stack Learning Management System inspired by platforms like Udemy, supporting multiple user roles (Student, Instructor, Admin).

Key Features:

- Built complete MERN stack architecture (separate frontend & backend)
- Designed RESTful APIs for users, courses, and enrollment
- Implemented OTP-based authentication with secure JWT(json web token)
- Role-Based Access Control (RBAC) for different user types

- Instructor dashboard for course creation (videos, thumbnails, tags)
- Student dashboard with enrollment and progress tracking
- Integrated Cloudinary for media uploads
- Implemented secure authentication using httpOnly cookies
- Handled real-world issues (CORS, API errors, deployment bugs)

Tech stack:

React.js, Node.js, Express.js, MongoDB, Mongoose, JWT, Cloudinary, Tailwind CSS

Authentication System (Zustand + JWT)

☞ **Frontend:** <https://vercel.com/sheryar7s-projects/auth-app-frontend>

☞ **GitHub (Frontend):** <https://github.com/Sheryar7/auth-app-frontend>

☞ **GitHub (Backend):** <https://github.com/Sheryar7/auth-app-backend>

Description:

Built a complete authentication system focusing on modern state management using Zustand instead of Redux.

Key Features:

- Signup/Login with JWT(json web token)
- OTP email verification system
- Forgot/Reset password flow
- Global state management using Zustand
- Protected routes and middleware handling

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
- **Frontend:** React.js, Next.js (App Router), Tailwind CSS, Redux Toolkit, Zustand, Responsive Web Design
- **Backend:** Node.js, Express.js, NestJS, Supabase, RESTful APIs, JWT Authentication, Middleware, RBAC, Retrieval-Augmented Generation (RAG), Vector Search, Embeddings
- **Databases:** PostgreSQL, MongoDB & Mongoose, SQLite & TypeORM, PostgreSQL (pgvector), Supabase, MongoDB & Mongoose, SQLite & TypeORM
- **Tools & Platforms:** Git, GitHub, Postman, Vercel, VS Code, AI-Assisted Workflows (Copilot, ChatGPT, Claude)

STRENGTHS

- **Rapid Self-Learner & Adaptable:** Proven track record of independently mastering modern stacks (MERN, Next.js 15, NestJS, TypeScript) and quickly adapting to production environments.
- **Full-Stack Systems Architecture:** Solid understanding of end-to-end web architectures—including RESTful API design, database modeling (Supabase/PostgreSQL, MongoDB), and modern server/client rendering patterns.
- **Debugging & Problem Solving:** Strong hands-on capability in isolating and resolving complex TypeScript, state management, and production backend/build issues efficiently.
- **AI and RAG(Retrieval-Augmented Generation) system integration:** Experienced in architecting Retrieval-Augmented Generation (RAG) pipelines, integrating vector databases (pgvector), generating LLM embeddings (Gemini API), and leveraging AI tools (Copilot, Claude) to build intelligent, context-aware web applications.
- **Clean & Scalable Code Base:** Focused on writing modular, maintainable, and reusable component/service structures adhering to design patterns (MVC, NestJS modules, DTO validations).

EDUCATION

Abasyn University, Peshawar
Bachelor of Software Engineering

Graduated: September 2023